

Everest Engineering College

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Assignment 2 , 2024

Level : BE First Semester

Subject: Calculus I

Due Date: Dec 22, 2024

Students are required to give their answers in their own words as far as practicable and copy from others is strictly Prohibited.

Attempt all the Questions:

1. Find y_n of $y = (ax - b)^m$.
2. Find y_n of $y = \frac{x^n}{x-1}$. [Hint: see similar example in example 30]
3. Find y_n of $y = \cos x$.
4. Find y_n of $y = \frac{1}{x^2+9}$. [**Hint: see similar example in example 28**]
5. Find y_n of $y = x^3 \cos x$.
6. Find y_n of $y = e^{ax} \log x$. [**Hint: see similar example in example 36**]
7. If $y = (\tan^{-1})^2$, show that $(x^2 + 1)y_2 + 2x(x^2 + 1)y_1 - 2 = 0$
8. If $y = a \cos(\log x) + b \sin(\log x)$, then prove that
 - a. $x^2 y_2 + x y_1 + y = 0$
 - b. $x^2 y_{n+2} + (2n + 1) x y_{n+1} + (n^2 + 1) y_n = 0$.[**Hint: see similar example in example 37**]
9. If $y = e^{a \tan^{-1} x}$, then prove that
 - a. $(1 + x^2) y_2 + (2x - a) y_1 = 0$
 - b. $(1 + x^2) y_{n+2} + (2nx + 2x - a) y_{n+1} + n(n + 1) y_n = 0$.
10. If $y^{1/m} + y^{-1/m} = 2x$, then prove that
 - a. $(x^2 - 1) y_1^2 - m^2 y^2 = 0$
 - b. $(x^2 - 1) y_{n+2} + (2n + 1) x y_{n+1} + (n^2 - m^2) y_n = 0$.[**Hint: see similar example in example 38**]
11. If $y = (x + \sqrt{1 + x^2})^m$, then prove that
 - a. $(1 + x^2) y_2 + x y_1 - m^2 y = 0$
 - b. $(1 + x^2) y_{n+2} + (2n + 1) x y_{n+1} + (n^2 - m^2) y_n = 0$.
12. If $y = \sin(m \sin^{-1} x)$, then prove that
 - a. $(1 - x^2) y_2 - x y_1 + m^2 y = 0$
 - b. $(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} + (m^2 - n^2) y_n = 0$.

13. If $y = \sin^{-1} x$, then prove that
- $(1 - x^2)y_2 - xy_1 = 0$
 - $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$.
 - $(y_{n+2})_0 = n^2(y_n)_0$. Then find the value of $(y_n)_0$.
- [Hint: see similar example in example 43]**
14. If $y = x^{n-1} \log x$, then prove that
- $xy_1 = (n - 1)y + x^{n-1}$
 - $xy_n = (n - 1)!$
- [Hint: $y_1 = (n - 1)x^{n-2} \log x + x^{n-1} \frac{1}{x} = (n - 1)x^{n-2} \log x + x^{n-2} = (n - 1)y \cdot \frac{1}{x} + x^{n-2}$
 $\Rightarrow xy_1 = (n - 1)y + x^{n-1}$, then we get (a) and for (b) apply Leibnitz rule upto (n-1)th order derivative]**
15. Verify Rolle's Theorem for
- $f(x) = x(x - 8)e^{\left(\frac{-x}{2}\right)}$ on $[0, 8]$
 - $f(x) = e^x(\sin x - \cos x)$ on $[\pi/4, 5\pi/4]$
 - $f(x) = (x - 2)(x - 3)(x - 4)$ on $[2, 4]$
 - $f(x) = \log\left(\frac{x^2 + ab}{(a+b)x}\right)$ on $[a, b]$, $a > 0$.
16. Verify Lagrange's Mean Value Theorem for
- $f(x) = e^x$ on $[1, 4]$
 - $f(x) = \log x$ on $[1, e]$
 - $f(x) = x + \frac{1}{x}$ on $[1/2, 2]$
 - $f(x) = x(x - 1)(x - 2)$ on $[0, 1/2]$
 - $f(x) = Ax^2 + Bx + C$, $A \neq 0$ on $[a, b]$
17. Discuss the applicability of Rolle's and Lagrange's Mean Value Theorems for the function $f(x) = 1 - (x - 1)^{(3/2)}$ on $[0, 2]$.
18. Assuming the validity of the expansion, Prove the following by using Malaurin's series:
- $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$
 - $e^{x/2} = 1 + \frac{x}{2} + \frac{x^2}{4 \cdot 2!} + \frac{x^3}{8 \cdot 3!} + \dots$
 - $\sin(e^x - 1) = x + \frac{x^2}{2} - \frac{5x^4}{24} + \dots$
 - $\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$
 - $e^x \log(1 + x) = x + \frac{x^2}{2!} + \frac{2x^3}{3!} + \frac{9x^5}{5!} + \dots$
19. Expand $f(x) = \log(1 + e^x)$ as far as the term x^4 by Maclaurin's series and hence the expansion of $\frac{e^x}{1 + e^x}$.
20. Examine the validity of Maclaurin's series for the function $f(x) = x^{\frac{5}{2}}$.

"ALL THE BEST."